cNODE: A Continuous-Time Neural ODE Transformer for Early Sepsis Prediction

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Abstract

Early prediction of sepsis from electronic health record is challenging because clinical time series are sparse, irregularly sampled, and heavily affected by missingness. We propose a continuous-time Transformer framework that explicitly models patient trajectories along clinical time. The model integrates missingness-aware token representations with neural ordinary differential equation (ODE)-based latent dynamics, allowing patient states to evolve continuously between observations without time discretization. This formulation enables more faithful modeling of irregular clinical events compared to conventional discrete-time architectures. We evaluate the approach on an ICU cohort in MIMIC-IV across multiple prediction horizons prior to sepsis onset. At the 24-hour prediction horizon, the model achieves an AUROC of 0.734 and an AUPRC of 0.769, outperforming existing methods. These results demonstrate that combining Transformer representations with continuous-time latent dynamics provides an effective approach to modeling irregular clinical time series and improving early sepsis risk prediction.

1 Introduction

Sepsis is a life-threatening organ dysfunction caused by a dysregulated host response to infection and can lead to organ failure and morbidity.[1,2] Prior works[3,4,5] indicate that delaying recognition of sepsis patients causes a substantial increase in adjusted mortality. Specifically, Kumar et al.[6] noted an hourly mortality increase of 3.6–9.9% for patients in septic shock, demonstrating that timely identification and intervention are crucial in intensive care settings.[7] As a result, early sepsis prediction has become an important research problem in clinical machine learning.

Traditional sepsis screening scores, such as SIRS,[8] qSOFA, and SOFA,[2] remain widely used in clinical practice because of their simplicity and interpretability.[9] In parallel, numerous machine learning and deep learning models have been developed to leverage electronic health record data for earlier and more accurate sepsis prediction.[10] Nevertheless, early sepsis prediction remains difficult because ICU time-series data are inherently irregular, sparse, and incomplete.[11] Physiological variables are recorded according to clinical need rather than at fixed analytic intervals, resulting in unevenly sampled trajectories.[12] To make such data usable for standard sequence models, many pipelines rely on aggregation, forward filling, or imputation. However, these preprocessing strategies are often not standardized across studies,[13] and uncertainty from missing-value imputation can propagate to downstream prediction results.[14] More importantly, such processing may obscure the true temporal evolution of patient states and weaken clinically meaningful timing information. This issue is particularly relevant for sepsis, where both short-term physiological changes and the elapsed time between observations may carry predictive value.[15]

These challenges motivate models that can represent patient trajectories in continuous time while preserving irregular temporal structure. Although many existing sequence models operate on discretized time grids,[15,13,16,17] where they often use fixed hourly features, the exact timing between clinical observations may itself carry useful information for early sepsis prediction. Neural ODEs provide a principled framework for modeling latent state evolution in continuous time and are therefore well suited to irregularly sampled ICU time series. At the same time, Transformer-based architectures are effective at learning complex interactions among multivariate clinical variables.[15,17]

Continuous-time and time-aware Transformers have recently been applied to irregular clinical data in several ways. Time-aware models such as T-LSTM,[18] GRU-D,[19] and RAPT[20] incorporate elapsed time or decay terms while still operating on aggregated or fixed time steps. Other approaches combine attention with neural differential equations: ContiFormer[21] makes the attention operation itself continuous, while the continuous patient-state attention model[22] applies a Neural ODE to regularize the series before attention. Continuous-depth ODE Transformers[23] instead treat the ODE time variable as network depth with time-varying weights.

Motivated by these complementary strengths, we develop a Neural ODE-based Transformer framework for early

sepsis prediction that combines missingness-aware temporal representations with continuous-time latent dynamics. The contributions of our work are:

- We introduce cNODE, which uses a Transformer block as the vector field of a Neural ODE to evolve the full latent token sequence, including a global classification token, in continuous physical time, with the network weights held fixed and elapsed time injected into the latent state, paired with missingness-aware tokenization for early sepsis prediction.
- We provide empirical analysis of the proposed design to examine the value of continuous-time modeling and missingness-aware temporal representation for clinical time-series prediction.
- Our novelty is to use a Transformer block as the vector field of a Neural ODE, evolving the entire latent token sequence over real elapsed time for early sepsis prediction, with network weights held fixed and physical time injected into the latent state. This formulation departs from prior continuous-time Transformers that instead make attention itself continuous or treat the ODE's time variable as network depth, paired with missingness-aware tokenization.

2 Materials and Methods

This section describes our proposed Continuous-time Neural ODE Transformer method in early sepsis prediction. We first detail the dataset, the problem formulation and our data representation and tokenization strategy, followed by a comprehensive description of our method.

2.1 Dataset and problem setup

We extract the study cohort from MIMIC-IV[24,25] dataset, a publicly accessible electronic health record dataset. Eligible participants were adult patients with ICU length of stay exceeding 36 hours. For each patient, we select their first ICU stay. The study considers ten candidate dynamic clinical variables: heart rate (HR), respiratory rate (RR), oximetry saturation (SpO_2), mean arterial pressure (MAP), temperature, white blood cell count (WBC), creatinine, platelet count, lactate, and bilirubin. After statistical screening and missingness analysis (Section 3.1), eight are retained for modeling; lactate and bilirubin are excluded owing to weak univariate association and high missingness.

MIMIC-IV is a single-center database derived from a large tertiary academic medical center (Beth Israel Deaconess Medical Center, Boston, MA) with a mature, fully electronic clinical information system. ICU physiological measurements are captured at the bedside by the clinical information system and laboratory results by the hospital laboratory system during routine clinical care. Because measurements are recorded according to clinical need rather than on a fixed schedule, the resulting time series are irregularly sampled and exhibit variable missingness across variables (Section 3.1.2). To improve the reliability of the extracted measurements, we harmonized units and removed physiologically implausible values using per-variable plausibility ranges before binning and aggregation. Table 1 summarizes the baseline characteristics of the evaluated cohort by sepsis status. Sepsis labels and onset times are derived from the Sepsis-3 cohort in `mimiciv_derived.sepsis3` table[24,26] where a patient is defined as a case$(y = 1)$ if it had a documented Sepsis-3 onset time o, and a control $(y = 0)$ otherwise. We define an anchor time $\mathrm{t}_{\mathrm{anchor}}$ for each stay to construct comparable retrospective observation windows. For cases, the anchor time was set to the sepsis onset time $\mathrm{t}_{\mathrm{anchor}} = \mathrm{o}$, with $\mathrm{o} \in [\mathrm{t}_{\mathrm{in}}, \mathrm{t}_{\mathrm{end}}]$. For controls, we match the time-in-stay distribution of cases by first computing case offsets by calculating $\Delta_{\mathrm{case}} = \mathrm{t}_{\mathrm{anchor,case}} - \mathrm{t}_{\mathrm{in}}$, then sample Δ for each control from the empirical distribution of Δ_{case}. The control index time is defined as $\mathrm{t}_{\mathrm{anchor}} = \min(\mathrm{t}_{\mathrm{in}} + \Delta, \mathrm{t}_{\mathrm{end}})$.

The early sepsis prediction problem is defined as estimating the probability that an ICU stay will develop sepsis at a prediction time $\mathrm{t}_{\mathrm{pred}}$, where $\mathrm{t}_{\mathrm{pred}}$ is defined h hours prior to anchor time $\mathrm{t}_{\mathrm{anchor}}$:

$$\mathrm{t}_{\mathrm{pred}} = \mathrm{t}_{\mathrm{anchor}} - \mathrm{h}$$

For each prediction horizon h, the model input is a fixed context window of 24 hours $[\mathrm{t}_{\mathrm{pred}} - 24, \mathrm{t}_{\mathrm{pred}}]$. We exclude stays where $\mathrm{t}_{\mathrm{anchor}} \leq \mathrm{t}_{\mathrm{in}}$ and where $\mathrm{t}_{\mathrm{anchor}} > \mathrm{t}_{\mathrm{end}}$.

Table 1: Baseline characteristics of the study cohort

Characteristic	Overall (N=5073)	Control (n=2405)	Sepsis (n=2668)	p-value
No. of ICU stays	5073	2405	2668	
Age, years, mean ± SD	63.0 ± 16.9	63.0 ± 16.9	63.0 ± 16.9	0.911
Male sex, n (%)	2745 (54.1)	1296 (53.9)	1449 (54.3)	0.785
Race, n (%)				< 0.001
White	3135 (61.8)	1557 (64.7)	1578 (59.1)	
Black	474 (9.3)	237 (9.9)	237 (8.9)	
Hispanic/Latino	202 (4.0)	98 (4.1)	104 (3.9)	
Asian	158 (3.1)	74 (3.1)	84 (3.1)	
Other/Unknown	1104 (21.8)	439 (18.3)	665 (24.9)	
First ICU care unit, n (%)				< 0.001
MICU	714 (14.1)	303 (12.6)	411 (15.4)	
MICU/SICU	535 (10.5)	239 (9.9)	296 (11.1)	
SICU	855 (16.9)	369 (15.3)	486 (18.2)	
CVICU	658 (13.0)	371 (15.4)	287 (10.8)	
CCU	585 (11.5)	328 (13.6)	257 (9.6)	
TSICU	714 (14.1)	303 (12.6)	411 (15.4)	
Neuro ICU/stepdown	969 (19.1)	483 (20.1)	486 (18.2)	
Other	43 (0.8)	9 (0.4)	34 (1.3)	
Admission type, n (%)				< 0.001
Emergency	2692 (53.1)	1196 (49.7)	1496 (56.1)	
Urgent	929 (18.3)	379 (15.8)	550 (20.6)	
Observation	928 (18.3)	516 (21.5)	412 (15.4)	
Elective	91 (1.8)	48 (2.0)	43 (1.6)	
Surgical same-day	433 (8.5)	266 (11.1)	167 (6.3)	
SOFA (first 24h), median [IQR]	3.0 [1.0, 5.0]	2.0 [1.0, 4.0]	3.0 [2.0, 5.0]	< 0.001
Charlson comorbidity index, median [IQR]	5.0 [3.0, 7.0]	4.0 [2.0, 7.0]	5.0 [3.0, 7.0]	0.066
ICU length of stay, days, median [IQR]	4.1 [2.3, 8.9]	2.6 [1.9, 3.9]	7.8 [4.2, 14.0]	< 0.001
In-hospital mortality, n (%)	572 (11.3)	125 (5.2)	447 (16.8)	< 0.001

2.2. Data representation and tokenization

We denote the cohort as $\mathcal{D} = \{(X_i, y_i)\}_{i=1}^{N}$, where i indexes ICU stays, N is the number of stays, $y_i \in \{0, 1\}$ is the sepsis label. F is the number of dynamic features used for modeling ($F = 8$). For stay i, raw measurements are recorded at irregular times. We standardize timestamps by mapping each measurement time t to a 5-minute bin, denoted by $\tilde{t} = \lfloor t \rfloor_{5\text{min}}$. If multiple measurements of the same feature fall into the same bin, we aggregate them by their mean. Importantly, we retain only bins containing at least one measurement rather than constructing a fully dense 5-minute grid, so the time interval between two consecutive bins is not always the same. We represent the binned data for each stay i by a value matrix $X_i^\top = \begin{bmatrix} x_{i,1} & x_{i,2} & \cdots & x_{i,T_i} \end{bmatrix} \in \mathbb{R}^{F \times T_i}$ where T_i is the number of retained time bins for stay i and each column vector $x_{i,j} \in \mathbb{R}^F$ is the feature vector at the jth time bin. $M_i^\top = \begin{bmatrix} m_{i,1} & m_{i,2} & \cdots & m_{i,T_i} \end{bmatrix} \in \{0, 1\}^{F \times T_i}$ is the missing mask matrix generated from X_i, where each column vector $m_{i,j} \in \{0, 1\}^F$ indicates whether each feature is observed at the jth time bin. For each feature, we further define a per-feature gap vector $\Delta\tilde{t}_{i,j} \in \mathbb{R}^F$, whose entries give the elapsed time since that feature was last observed, and use the transformed gap feature $\log(1 + \Delta\tilde{t}_{i,j}) \in \mathbb{R}^F$ as part of the model input. For each dynamic time step j, we construct a token from four components: the masked observed value $X \odot M$, the observation mask M to capture informative missingness, the irregular time gap feature $\log(1 + \Delta\tilde{t}_{i,j})$, and forward-filled last observed value X_{last}. Formally, for each stay i and retained time step j, we have:

$$z_{i,j}^{\text{raw}} = \left[x_{i,j} \odot m_{i,j},\ m_{i,j},\ \log(1 + \Delta\tilde{t}_{i,j}),\ x_{i,j}^{\text{last}} \right] \in \mathbb{R}^{4F}$$

2.3. Continuous-time Neural ODE Transformer

In this study, we propose cNODE, an irregular-time Neural ODE Transformer, designed to evolve patient representations over time using clinical tokens for early sepsis prediction. Our study first projects raw dynamic token $z_{i,j}^{\mathrm{raw}}$ into a d-dimensional latent space

$$e_{i,j}^{\mathrm{dyn}} = \phi_{\mathrm{dyn}}(z_{i,j}^{\mathrm{raw}}) \in \mathbb{R}^d,$$

where ϕ_{dyn} denotes a learnable two-layer feed-forward projection with GELU activation. We then compute a time embedding from the corresponding timestamp $t_{i,j}$ using a sinusoidal time embedding followed by a learnable linear projection, and add it to the projected token representation:

$$e_{i,j}^{0} = e_{i,j}^{\mathrm{dyn}} + e_{i,j}^{\mathrm{time}}.$$

Inspired by BERT,[27] we prepend a learned classification token to the temporal token sequence. The resulting input sequence is $H_i^0 = \left[h_{\mathrm{cls}}, e_{i,1}^0, e_{i,2}^0, \ldots, e_{i,T_i}^0\right]$.

Pre-ODE fusion block. Before continuous-time evolution, we apply a Transformer-style fusion block to the input sequence. The block consists of self-attention (Attn) and a position-wise feed-forward network (FFN) with residual connections and layer normalization (LN). Formally, the fused representation is written as

$$H_i(0) = \mathrm{FusionBlock}(H_i^0).$$

Continuous-time token dynamics. Given the initial sequence representation $H_i(0)$ for stay i, which contains the learned [CLS] token and all temporal tokens, we evolve the full latent token sequence through continuous-time dynamics. This formulation differs from continuous-depth Neural ODE Transformers such as Tong et al.,[23] in which time corresponds to continuous network depth and the Transformer weights are parameterized as time-varying functions of the layer index. In our setting, the dynamics are governed by a Transformer-style vector field f_θ, and the ODE is solved independently for each patient over their clinical timestamps. Specifically, the latent dynamics are defined as:

$$\frac{dH_i(t)}{dt} = f_\theta(t, H_i(t)),$$

where the vector field f_θ is parameterized by layer normalization, multi-head self-attention, and a position-wise feed-forward network with learnable residual scaling parameters. For a latent sequence state $H(t) \in \mathbb{R}^{L\times d}$ containing $L = T_i + 1$ tokens (the [CLS] token and the T_i temporal tokens), the vector field is defined as:

$$f_\theta(t, H(t)) = \lambda_a \,\mathrm{Attn}_\theta(\mathrm{LN}(H_\psi(t))) + \lambda_b \,\mathrm{FFN}_\theta(\mathrm{LN}(H_\psi(t)))\,,$$

where λ_a and λ_b are learnable scaling parameters. Time conditioning is achieved by mapping the scalar ODE time t into a sinusoidal embedding, denoted as $\psi(t) \in \mathbb{R}^d$. Prior to the attention and feed-forward operations, this time embedding is broadcast across all L tokens to yield the time-conditioned state $H_\psi(t)$, defined as:

$$H_\psi(t) = H(t) + \mathbf{1}_L \psi(t)^\top,$$

where $\mathbf{1}_L \in \mathbb{R}^{L\times 1}$ is a column vector of ones. In this way, the ODE derivative is governed by a continuous-time Transformer operator, so that f_θ depends on both the instantaneous hidden state $H(t)$ and the actual physical time t elapsed since the beginning of the observation window. This mechanism differs from that of Tong et al.,[23] who treat time as continuous network depth and define the vector field as $\frac{dH(t)}{dt} = \mathrm{Attn}_{W(t)}(H(t)) + \mathrm{FFN}_{W(t)}(H(t))$, where $W(t)$ denotes time-varying weight matrices. In contrast, our formulation keeps the model weights θ fixed and injects physical time directly into the evolving latent state through the additive embedding $H_\psi(t)$. This design allows the model to follow true irregular clinical time intervals and better align risk prediction with the actual prediction time. Formally, for stay i, the hidden state at time t_k is computed by integrating from the preceding valid observation time t_{k-1}:

$$H_i\left(t_k\right) = H_i\left(t_{k-1}\right) + \int_{t_{k-1}}^{t_k} f_\theta\left(\tau, H_i(\tau)\right)\, d\tau.$$

2.4. Prediction head and optimization

After continuous-time evolution, we obtain a final latent sequence $H_i^{\text{final}} = \left[x_i^{\text{cls}}, X_i^{\text{time}}\right]$ for stay i. Here $x_i^{\text{cls}} \in \mathbb{R}^d$ is the evolved [CLS] token to capture global trajectory information and $X_i^{\text{time}} \in \mathbb{R}^{T_i \times d}$ contains the evolved representations for each individual time point.

Temporal pooling and risk prediction. Rather than simply averaging over time, we let the model learn which time points matter most for predicting sepsis. Following the attention-pooling approach ,[28] we compute a scalar importance score for each temporal token:

$$u_{i,j} = a_\phi\left(x_{i,j}^{\text{time}}\right), \qquad \alpha_{i,j} = \frac{\exp(u_{i,j})}{\sum_{k=1}^{T_i} \exp(u_{i,k})},$$

and obtain a weighted sum $h_i^{\text{pool}} = \sum_{j=1}^{T_i} \alpha_{i,j}\, x_{i,j}^{\text{time}}$ which helps the model focus on sudden deterioration. The pooled temporal representation is then combined with the evolved [CLS] token by

$$h_i = \frac{1}{2}\left(x_i^{\text{cls}} + h_i^{\text{pool}}\right).$$

Finally, a linear layer maps h_i to a sepsis risk score:

$$\hat{r}_i = W_{\text{out}} h_i + b_{\text{out}}, \qquad \hat{y}_i = \sigma(\hat{r}_i).$$

Loss and optimization. The model is trained with binary cross-entropy loss. All parameters are optimized using AdamW, with gradient clipping for training stability and validation AUROC for model selection.

3. Experiments and results

In this section, we first present a feature-level analysis, including statistical feature screening and missingness assessment. We then describe the experimental setup and compare the proposed method with baseline models to evaluate its effectiveness. Finally, we conduct ablation studies to further examine the contributions of individual components of the proposed framework.

3.1 Feature Analysis

3.1.1 Statistical Feature Screening and Selection

We screened all ten candidate variables, and the results are summarized in Table 2. Each candidate was first assessed individually using the Mann–Whitney U test(MW) and univariate logistic regression, from which we report coefficient p-values, AUROC, and AUPRC. We then fitted a multivariable logistic regression(MLR) model including all candidate features to evaluate their adjusted associations with sepsis and report the corresponding coefficient p-values. Finally, L1-regularized logistic regression was applied as an embedded feature-selection method to identify a sparse subset of informative predictors across prediction horizons. Temperature and respiratory rate showed the strongest and most consistent univariate associations with sepsis across all prediction horizons. White blood cell count and platelet count also demonstrated stable associations, particularly at shorter horizons, while MAP and HR provided consistently available and clinically relevant vital-sign information. Creatinine and SpO_2 were retained as complementary predictors because they remained clinically meaningful and contributed additional information in multivariable analysis. In contrast, bilirubin and lactate were not prioritized as core predictors because of their weaker overall performance and substantially higher missingness.

3.1.2 Feature Missingness

We quantified feature missingness across prediction horizons. The median missing rate of 10 features is 10.1%. At 24 hour horizon, Bilirubin(69.9%) and lactate(59.3%) showed the highest missingness across all horizons, indicating

Table 2: Feature Screening Results

	24h horizon					18h horizon					12h horizon				
Feature	AUROC	AUPRC	Univ. p	MW p	MLR p	AUROC	AUPRC	Univ. p	MW p	MLR p	AUROC	AUPRC	Univ. p	MW p	MLR p
Temperature	0.61	0.66	4.93E-15	8.74E-17	9.09E-12	0.60	0.65	3.26E-18	6.26E-20	1.59E-14	0.59	0.63	2.05E-18	8.12E-21	1.15E-11
RR	0.59	0.63	4.39E-14	8.16E-14	2.18E-10	0.60	0.64	4.18E-19	6.64E-19	2.31E-15	0.59	0.64	2.76E-24	4.62E-22	6.47E-16
WBC	0.55	0.59	6.06E-06	1.17E-05	1.80E-03	0.55	0.58	4.87E-06	6.50E-07	1.02E-02	0.56	0.58	1.74E-08	4.32E-12	3.29E-03
MAP	0.54	0.57	1.93E-03	1.51E-03	5.67E-03	0.55	0.58	1.83E-05	5.15E-05	1.88E-04	0.53	0.55	2.31E-04	7.13E-05	1.69E-02
Platelet	0.54	0.58	1.09E-03	3.47E-03	8.78E-02	0.55	0.59	1.80E-05	4.93E-05	9.30E-03	0.56	0.60	3.43E-11	3.66E-11	1.49E-05
HR	0.54	0.58	3.19E-03	5.19E-03	5.42E-02	0.54	0.57	3.02E-04	6.05E-04	6.29E-02	0.55	0.59	9.17E-09	1.93E-07	9.82E-01
SpO_2	0.51	0.55	7.28E-01	4.51E-01	5.92E-02	0.48	0.53	8.82E-01	6.00E-01	2.98E-02	0.51	0.55	7.56E-01	2.88E-01	3.60E-03
Creatinine	0.51	0.55	1.34E-01	8.28E-02	6.35E-01	0.52	0.56	6.22E-02	4.16E-02	3.17E-01	0.52	0.55	5.80E-03	1.65E-02	8.54E-02
Bilirubin	0.49	0.55	5.46E-02	9.39E-01	1.36E-02	0.50	0.55	1.86E-02	8.80E-01	1.73E-03	0.49	0.54	1.91E-01	1.89E-01	2.81E-02
Lactate	0.49	0.55	7.13E-02	7.61E-01	2.57E-01	0.48	0.53	1.84E-01	5.24E-01	3.47E-01	0.48	0.52	7.22E-01	4.49E-02	7.02E-01

limited reliability as core predictors, whereas HR, RR, SpO_2, and MAP had very low stay-level missingness and broad coverage. Temperature was available in most stays despite less frequent temporal sampling, while WBC, platelet count, and creatinine showed higher missingness than the vital signs, but were still observed in many ICU stays, providing complementary laboratory information. Combining this missingness analysis with the statistical screening in Section 3.1.1, we retained eight of the ten candidate variables for modeling: WBC, temperature, creatinine, platelet count, HR, RR, SpO_2, and MAP. Bilirubin and lactate were excluded from the final feature set because of their weak univariate performance and substantially higher missingness.

3.2 Experiment Setup

We conduct experiments on the MIMIC-IV ICU dataset.[24,25] The cohort was split into training (72%), validation (8%), and test (20%) sets. Unless otherwise stated, hyperparameters were selected on the validation set, and final model performance was evaluated on the held-out test set. Table 3 summarizes the selected dynamic features across prediction horizons (24h/18h/12h). For each feature, we report the number of ICU stays, the mean $\pm$ standard deviation in the sepsis-negative and sepsis-positive groups, the corresponding univariate p-values, and the percentage of missing values within the observation window. Each entry is presented as train/test, where the train statistics were computed from the combined training and validation sets. Overall, the train and test distributions were broadly similar across prediction horizons. Temperature and respiratory rate showed consistent group differences, while WBC, platelet count, HR, and MAP also exhibited separation in the training cohort. SpO_2 and creatinine showed weaker and less consistent univariate differences. Missingness was low to moderate for the vital signs (approximately 18%–23%) but substantially higher for laboratory variables, particularly WBC, creatinine, and platelet count (approximately 94%).

Baseline methods. Our study compares the proposed model with the following baseline methods:

- T-LSTM[18] : An LSTM model for irregular time intervals that uses time gaps to decay memory.
- GRU-D[19] : A GRU model that learns decay rates and imputes missing values during training.
- RAPT[20] : A Transformer model that uses time-gap-aware attention for irregular timestamps.

Evaluation metrics. We compare the model performance using the Area Under the Receiver Operating Characteristic curve (AUROC) and the Area Under the Precision-Recall Curve (AUPRC). Furthermore, we report Sensitivity (Sen) and Specificity (Spe) to explicitly quantify the model's ability to correctly identify true positive and true negative cases, respectively.

Implementation details. We trained the cNODE model on the Yale Bouchet cluster with 32GB memory and an NVIDIA RTX 5000 Ada GPU. The architecture was configured with a latent dimension of $d = 64$, 4 attention heads, and a hidden feed-forward dimension of 128. We compute continuous-time dynamics using the `dopri5` solver, setting relative and absolute tolerances to 10^{-3} and 10^{-4}, respectively. Models were trained using AdamW (batch size 64, learning rate 4e-4, weight decay 1e-4) for a maximum of 40 epochs. The learning rate was decayed by a factor of 0.5 upon validation plateau (patience 3, minimum LR 1e-6). Training terminates via early stopping if the validation AUROC does not improve by at least 1e-4 for 10 consecutive epochs. All runs were fixed to a random seed of 42.

Table 3: Summary statistics of selected dynamic features across prediction horizons.

	Sepsis Negative		Sepsis Positive		p-value		Missingness (%)	
Baseline characteristic	Train	Test	Train	Test	Train	Test	Train	Test
24h prediction horizon								
Number of ICU stays, n	1,015	277	1,178	292	–	–	–	–
White blood cell count (mean ± SD)	10.8 ± 4.7	11.8 ± 5.8	12.0 ± 5.6	11.8 ± 5.4	< 0.001	0.985	94.1	93.6
Temperature (mean ± SD)	36.8 ± 0.4	36.9 ± 0.4	37.0 ± 0.5	37.0 ± 0.5	< 0.001	0.001	76.6	76.9
Creatinine (mean ± SD)	1.2 ± 1.2	1.2 ± 1.1	1.2 ± 1.0	1.2 ± 1.3	0.311	0.425	93.7	93.3
Platelet count (mean ± SD)	205.8 ± 90.5	213.1 ± 103.5	221.3 ± 108.4	215.1 ± 106.0	< 0.001	0.831	94.0	93.5
Heart rate (mean ± SD)	82.9 ± 16.1	83.5 ± 15.5	85.1 ± 17.3	85.7 ± 16.4	0.002	0.101	18.1	18.9
Respiratory rate (mean ± SD)	18.6 ± 3.6	18.9 ± 3.8	20.0 ± 4.3	19.9 ± 4.0	< 0.001	0.002	19.2	20.7
SpO_2 (mean ± SD)	96.8 ± 2.1	96.5 ± 2.2	96.9 ± 2.2	97.0 ± 2.1	0.589	0.003	19.1	20.2
Mean arterial pressure (mean ± SD)	82.0 ± 12.4	82.6 ± 11.7	83.7 ± 12.6	84.1 ± 12.7	0.001	0.136	20.8	21.1
18h prediction horizon								
Number of ICU stays, n	1,341	347	1,547	385	–	–	–	–
White blood cell count (mean ± SD)	11.0 ± 5.1	11.9 ± 5.6	12.2 ± 5.6	11.9 ± 5.0	< 0.001	0.965	94.0	93.7
Temperature (mean ± SD)	36.8 ± 0.4	36.8 ± 0.5	37.0 ± 0.5	37.0 ± 0.5	< 0.001	< 0.001	77.0	77.2
Creatinine (mean ± SD)	1.2 ± 1.1	1.2 ± 1.1	1.1 ± 0.9	1.2 ± 1.2	0.119	0.978	93.7	93.5
Platelet count (mean ± SD)	207.8 ± 92.3	212.6 ± 97.6	226.7 ± 110.6	224.0 ± 108.8	< 0.001	0.168	93.9	93.6
Heart rate (mean ± SD)	82.8 ± 16.5	83.2 ± 16.1	85.0 ± 17.4	85.6 ± 16.8	< 0.001	0.056	18.4	18.9
Respiratory rate (mean ± SD)	18.5 ± 3.6	18.8 ± 3.7	20.0 ± 4.3	19.9 ± 4.3	< 0.001	< 0.001	19.7	20.8
SpO_2 (mean ± SD)	96.8 ± 2.1	96.5 ± 2.2	96.8 ± 2.2	96.8 ± 2.2	0.719	0.127	19.6	20.2
Mean arterial pressure (mean ± SD)	82.1 ± 12.4	82.8 ± 12.0	84.0 ± 12.7	84.6 ± 12.2	< 0.001	0.041	21.8	21.7
12h prediction horizon								
Number of ICU stays, n	1,858	482	2,140	527	–	–	–	–
White blood cell count (mean ± SD)	11.1 ± 5.5	11.6 ± 5.6	12.4 ± 5.8	11.7 ± 5.0	< 0.001	0.958	93.8	93.9
Temperature (mean ± SD)	36.8 ± 0.5	36.8 ± 0.5	37.0 ± 0.5	36.9 ± 0.6	< 0.001	< 0.001	77.1	77.0
Creatinine (mean ± SD)	1.2 ± 1.1	1.2 ± 1.1	1.1 ± 0.9	1.1 ± 1.0	0.012	0.243	93.5	93.8
Platelet count (mean ± SD)	205.3 ± 92.6	211.3 ± 95.3	232.5 ± 113.9	225.3 ± 107.1	< 0.001	0.051	93.7	93.9
Heart rate (mean ± SD)	82.7 ± 16.2	82.7 ± 16.1	85.6 ± 18.0	85.0 ± 16.8	< 0.001	0.024	18.8	19.3
Respiratory rate (mean ± SD)	18.6 ± 3.6	18.5 ± 3.8	20.0 ± 4.6	19.6 ± 4.2	< 0.001	< 0.001	20.0	21.0
SpO_2 (mean ± SD)	96.8 ± 2.1	96.7 ± 2.2	96.9 ± 2.3	96.8 ± 2.3	0.588	0.469	20.1	20.4
Mean arterial pressure (mean ± SD)	82.4 ± 12.6	82.9 ± 11.9	83.7 ± 12.5	84.8 ± 12.9	0.001	0.013	22.6	22.7

3.3 Experiment results and analysis

Table 4 presents the validation and test performance of the proposed cNODE model and the baseline methods across three prediction horizons ($h = 12, 18, 24$). Among the 24 reported metrics, cNODE achieves the best result in 19 cases and consistently delivers the strongest performance on the primary discrimination metrics, AUROC and AUPRC. These results indicate that cNODE provides stronger discriminative ability than the competing methods and remains robust across different prediction horizons.

The superior performance of cNODE is consistent with its model design. Specifically, the model evolves the latent token sequence continuously over each patient's real ICU timestamps rather than over a fixed regular grid, allowing it to better capture irregular temporal dynamics in the clinical trajectory. In addition, the Transformer-style ODE vector field further enhances representation learning by modeling interactions among temporal tokens and the global $[\mathrm{CLS}]$ token, providing a richer sequence representation than the RNN-based baselines.

We further observe that performance differences across horizons are moderate rather than dramatic. This is expected because the three horizon-specific tasks are derived from the same set of patient stays, with the main difference being the location of the prediction time before sepsis onset. For positive samples, the prediction time is defined as $t_{\mathrm{pred}} = t_{\mathrm{onset}} - h$, and each sample is represented by a backward-looking clinical context window ending at t_{pred}. As a result, adjacent horizons yield highly related input distributions, leading to only moderate variation in predictive performance.

Table 4: Validation and test performance at different prediction horizons. Bold/underline indicates the best/second.

Horizon	Model	Val				Test			
		AUROC	AUPRC	Sen	Spe	AUROC	AUPRC	Sen	Spe
$h = 24$	RAPT	$\underline{0.768 \pm 0.016}$	$\underline{0.841 \pm 0.011}$	$\underline{0.657 \pm 0.060}$	0.704 ± 0.069	$\underline{0.732 \pm 0.012}$	$\underline{0.752 \pm 0.013}$	$\mathbf{0.680 \pm 0.062}$	0.627 ± 0.091
	T-LSTM	0.688 ± 0.006	0.791 ± 0.008	0.582 ± 0.030	$\underline{0.742 \pm 0.026}$	0.673 ± 0.013	0.694 ± 0.014	0.568 ± 0.055	0.660 ± 0.047
	GRU-D	0.658 ± 0.006	0.755 ± 0.013	0.606 ± 0.044	0.638 ± 0.033	0.653 ± 0.034	0.653 ± 0.038	0.616 ± 0.063	0.628 ± 0.043
	cNODE	$\mathbf{0.807 \pm 0.005}$	$\mathbf{0.875 \pm 0.002}$	$\mathbf{0.657 \pm 0.077}$	$\mathbf{0.763 \pm 0.066}$	$\mathbf{0.753 \pm 0.004}$	$\mathbf{0.782 \pm 0.007}$	$\underline{0.641 \pm 0.094}$	$\mathbf{0.712 \pm 0.103}$
$h = 18$	RAPT	$\underline{0.756 \pm 0.006}$	$\underline{0.827 \pm 0.005}$	0.600 ± 0.095	$\mathbf{0.754 \pm 0.085}$	$\underline{0.739 \pm 0.005}$	$\underline{0.753 \pm 0.004}$	0.636 ± 0.096	0.690 ± 0.077
	T-LSTM	0.659 ± 0.008	0.752 ± 0.007	$\underline{0.661 \pm 0.025}$	0.611 ± 0.014	0.687 ± 0.006	0.680 ± 0.010	$\mathbf{0.669 \pm 0.008}$	0.611 ± 0.013
	GRU-D	0.681 ± 0.022	0.766 ± 0.014	$\mathbf{0.663 \pm 0.043}$	0.601 ± 0.024	0.676 ± 0.009	0.656 ± 0.005	$\underline{0.671 \pm 0.043}$	0.599 ± 0.012
	cNODE	$\mathbf{0.784 \pm 0.012}$	$\mathbf{0.853 \pm 0.003}$	0.655 ± 0.065	$\underline{0.751 \pm 0.053}$	$\mathbf{0.747 \pm 0.007}$	$\mathbf{0.763 \pm 0.010}$	0.654 ± 0.066	$\mathbf{0.695 \pm 0.084}$
$h = 12$	RAPT	$\underline{0.735 \pm 0.007}$	$\underline{0.792 \pm 0.008}$	0.653 ± 0.011	$\mathbf{0.669 \pm 0.033}$	$\underline{0.715 \pm 0.006}$	$\underline{0.723 \pm 0.004}$	0.668 ± 0.043	0.632 ± 0.060
	T-LSTM	0.648 ± 0.016	0.721 ± 0.015	$\underline{0.671 \pm 0.041}$	0.550 ± 0.050	0.686 ± 0.001	0.687 ± 0.002	0.678 ± 0.046	0.595 ± 0.030
	GRU-D	0.652 ± 0.007	0.714 ± 0.001	$\mathbf{0.679 \pm 0.010}$	0.523 ± 0.019	0.701 ± 0.007	0.692 ± 0.005	$\mathbf{0.707 \pm 0.004}$	0.585 ± 0.020
	cNODE	$\mathbf{0.753 \pm 0.007}$	$\mathbf{0.817 \pm 0.003}$	0.645 ± 0.163	$\underline{0.683 \pm 0.141}$	$\mathbf{0.752 \pm 0.007}$	$\mathbf{0.757 \pm 0.004}$	$\underline{0.689 \pm 0.164}$	$\mathbf{0.648 \pm 0.168}$

3.4 Ablation study

We perform ablation studies to examine the contribution of each component of cNODE using the following variants:

- cNODE w/o ODE block: removes the continuous-time ODE module.
- cNODE w/o ODE-time input: removes $\psi(t)^\top$ from the ODE vector field while preserving the ODE block.
- cNODE w/o missingness mask: removes the observation mask M from the input
- cNODE w/o time-gap: removes the irregular time gap feature $\log(1 + \Delta\tilde{t}_{i,j})$
- cNODE with mean pooling: replace the final layer aggregation with the mean pooling
- cNODE with RK4 solver: use a fourth-order Runge–Kutta (RK4) solver to integrate the ODE

The ablation results in Table 5 show that removing the ODE block reduces both AUROC and AUPRC, indicating that the continuous-time latent evolution contributes to the overall discriminative performance of cNODE. In contrast, removing the ODE-time input leads to a slight decrease in AUPRC and a marginal change in AUROC. These results suggest that the ODE-time input has a relatively limited effect on ranking performance, but influences the operating characteristics of the model, particularly the balance between sensitivity and specificity.

Table 5: Ablation study results. Bold/underline indicates the best/second.

Variant	Val AUROC	Val AUPRC	Test AUROC	Test AUPRC
Full model	$\mathbf{0.807 \pm 0.005}$	$\underline{0.875 \pm 0.002}$	$\underline{0.753 \pm 0.004}$	$\mathbf{0.782 \pm 0.007}$
w/o ODE block	0.778 ± 0.009	0.853 ± 0.004	0.739 ± 0.013	0.757 ± 0.015
w/o ODE-time	0.791 ± 0.015	0.864 ± 0.009	0.737 ± 0.006	0.767 ± 0.005
w/o missingness mask	0.804 ± 0.003	$\underline{0.875 \pm 0.006}$	0.751 ± 0.006	0.776 ± 0.005
w/o time-gap	0.796 ± 0.020	0.870 ± 0.013	0.748 ± 0.005	0.775 ± 0.004
with mean pooling	$\underline{0.806 \pm 0.006}$	0.874 ± 0.001	0.749 ± 0.004	0.779 ± 0.003
with RK4 solver	$\underline{0.806 \pm 0.004}$	$\mathbf{0.876 \pm 0.002}$	$\mathbf{0.754 \pm 0.003}$	$\underline{0.781 \pm 0.009}$

4. Discussion

In this study, we proposed cNODE, a continuous-time Neural ODE Transformer for early sepsis prediction from sparse and irregular clinical time series. Across multiple prediction horizons, cNODE achieved consistently strong performance and outperformed the baseline models on the primary discrimination metrics, AUROC and AUPRC. These findings suggest that explicitly modeling patient trajectories in continuous time is beneficial for early sepsis prediction, particularly in ICU settings where measurements are asynchronous, sparse, and frequently missing.

The experimental results indicate that an important advantage of cNODE comes from its continuous-time latent evolution, the model evolves the latent token sequence across clinical timestamps, which may better preserve the temporal

structure of deterioration patterns preceding sepsis onset. In addition, the Transformer-based vector field enables interactions among temporal tokens and the global classification token during continuous-time evolution, supporting richer representation learning from multivariate clinical trajectories.

The ablation study further supports the importance of the ODE component. Removing the ODE block reduced both AUROC and AUPRC, indicating that continuous-time latent dynamics contribute to the overall discriminative performance of the model. By contrast, removing the ODE-time input produced only a marginal change in AUROC and a decrease in AUPRC, while increasing sensitivity. This suggests that explicit ODE-time conditioning has a more limited effect on ranking performance, but may influence the model's operating characteristics, particularly the balance between sensitivity and specificity.

This study has several limitations. First, static covariates and treatment-related information were not included, so the model only captures part of the clinical context available in ICU care. Second, the study was evaluated on a single MIMIC-IV-derived cohort, and external validation is needed to assess generalizability. Future work will extend the framework to multimodal clinical inputs, incorporate static and intervention-related variables, and evaluate performance on external cohorts. Third, the cohort originates from a single, well-resourced academic medical center with a mature clinical information system, so measurement completeness and case mix may differ in community or lower-resource ICUs; in addition, sepsis labels are derived algorithmically from the Sepsis-3 criteria rather than by manual chart adjudication.

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